ALVIN HANDOKO

0449710223 | alvinvarianto18@gmail.com | Perth, WA
https://alvinvh.github.io/alvins-page/

SOFTWARE DEVELOPER PROFILE:

- Master of IT (Software Engineering) graduate of Charles Darwin University with strong skills in Front-end Development, Back-end Development, Software Testing, Algorithms, Databases, and Machine Learning.
- Proficient in Python, C#, AngularJS, TypeScript, JavaScript, HTML, CSS, SQL, Flask, Django, and FastAPI for front-end development and back-end development during my final year project, coding competition project, and placement project.
- I am currently working as a Junior Programmer at Pro AV Solutions WA, where I have delivered multiple Audio Visual integration and control-system programming projects across education, corporate, and internal operational environments. In addition to AV programming, I design and develop web applications, user interfaces, and API-driven automation tools that streamline internal workflows and improve operational accuracy. I actively identify process gaps and build practical software solutions that enhance efficiency, reduce manual workload, and support both technical teams and daily business operations.

KEY SOFTWARE DEVELOPMENT SKILLS:

- **Programming Languages:** Python, C#, JavaScript, CSS, SQL
- **Front-end Development:** Experience in using AngularJS, JavaScript, HTML, CSS, Bootstrap, and Figma for front-end UI/UX design to develop web applications including WasteAI and Telemedicare in my final year project .
- **Back-end Development:** Skilled in Python, C#, Django, MySQL, and FastAPI to perform developed API for WasteAI and Telemedicare during my final year project.
- **Data Structures**: Experience in preparing data frames for machine learning such as handling missing value, normalization, augmentation, and label encoding using Pandas library.
- **Machine Learning**: Skilled in implementing multiple machine learning algorithms such as YOLO, Linear Regression, and CNN for WasteAI model and using tools such as TensorFlow, Ultralytics, and PyTorch.
- **Software Testing:** Experience in using Pytest, Robot Framework, OWASP ZAP, Burp Suite, and Metasploit to ensure the functionality and security of the WasteAI and Telemedicare for my final year project.
- **Version Control:** Experience in using Git and Github to manage version control, collaborative development, and consistently utilized it across all projects including my internship.
- **Technical Writing & Documentation:** Experience in writing code documentation and process guide for my internship and during university projects.

QUALIFICATIONS & CERTIFICATIONS:

- **CompTIA A+ -** CompTIA, completed January 2025
- **Master of IT Software Engineering –** Charles Darwin University, 2024
- **Bachelor of IT Network Security –** TAFE NSW, 2022

PROFESSIONAL EXPERIENCE

JUNIOR PROGRAMMER / SERVICE TECHNICIAN, PRO AV SOLUTIONS WA, PERTH WA, AUSTRALIA, JULY 2025 – PRESENT

Responsibilities:

- Configure and deploy professional AV systems in line with project design specifications, ensuring optimal performance and reliability.
- Develop, program, and automate AV device workflows, maintaining clean, version-controlled codebases through GitHub.
- Commissioning activities post-installation, validating system functionality, resolving integration issues, and delivering fully operational solutions to clients.
- Provide ongoing maintenance, diagnostics, and troubleshooting for client AV environments, ensuring minimal downtime and high service quality.
- Build internal automation tools and API-driven integrations to streamline workflows and enhance ticketing system efficiency.

SOFTWARE ENGINEER INTERNSHIP, DATA ENGINEERING PTY LTD, PERTH WA, AUSTRALIA, APRIL 2025 – JULY 2025

Responsibilities:

- Design a custom website for clients which include UI of the website using Figma to ensure the design is met client's and manager's standard.
- Develop custom Odoo module website using JavaScript, HTML, SCSS, PostgreSQL and Python that include developing the front-end of the website, creating back-end logic, database integration, and comprehensive functional testing.
- Proactively develop and manage documentation about the application and participate in weekly meeting to showcase weekly milestone.

SOFTWARE DEVELOPER, MOTHER THERESA CATHOLIC SCHOOL, ZUCCOLI NT, AUSTRALIA, FEBRUARY – JUNE 2024

Responsibilities:

- Designed a game for primary school children that teaches kindness, created the animation, sprites, content, and UI of the game using Figma and Unity to ensure that the game is entertaining for children while successfully reach the project's goal.
- Assisted project manager to ensure the development progress met the client's expectations, including but not limited to scheduling meetings, preparing meeting materials, and ensuring all team members reach their weekly milestones.
- Developed applications using Unity, C#, and .NET, including developing the front-end of the game, creating logic and game systems, and executing functional testing on all features.
- Created documentation about the application to ensure that the next team will be able to develop further or add new games to the application.

KEY IT PROJECTS:

PROJECT NAME: Southern Cross University New Campus (Pro AV Solutions WA)

Objective: The Southern Cross University Perth Campus project delivered a fully integrated AV ecosystem across a new multi-space facility, including 8 classrooms, 2 meeting rooms, a kitchen area, and a foyer space. The objective was to implement a seamless, user-friendly AV experience where most functions are automated, reducing the need for manual control and ensuring consistent performance across all rooms.

Key Tasks:

1. Configured and assigned AV devices according to detailed design documentation, ensuring accurate deployment across all teaching and meeting spaces.
2. Developed comprehensive CMDB documentation, mapping all AV assets, configurations, and network details for long-term maintainability.
3. Programmed Extron control systems in line with the programming scope, including: Automated display power control, Input switching for content sources, USB routing based on room mode, Audio volume control, Device monitoring and status reporting. These functions were achieved through integration of device modules and APIs.
4. Commissioned all AV systems post-installation, validating functionality, resolving issues, and ensuring each room met operational requirements.
5. Delivered training sessions for key university staff, covering system operation, automated features, and troubleshooting procedures.
6. Maintained full project documentation and version control using GitHub to ensure traceability and clean handover.

Outcome: The Southern Cross University Perth Campus AV integration project was successfully delivered on time and within budget, while maintaining a high standard of quality across all teaching and meeting spaces. Under the supervision of the Programming Manager, the programming work achieved an excellent result, with all systems performing reliably and meeting the university's operational expectations. The control system programming was developed using Python, leveraging Extron Global Scripter modules and device APIs to implement automation, monitoring, and user-friendly control logic. The final solution provided a consistent, intuitive experience across the campus, significantly reducing the need for manual intervention and ensuring long-term maintainability.

PROJECT NAME: Pro AV Internal integrations (Pro AV Solutions WA)

Objective: The Pro AV Internal Integrations project focused on improving operational efficiency by automating two major internal processes: generating Vimeo streaming links for Metropolitan Cemetery Board livestream funeral services and producing structured reporting dashboards from ConnectWise data. Previously, staff manually searched schedules and created streaming links, but the new software allows users to simply enter the name of the deceased and instantly retrieve the correct pre-generated Vimeo link. In addition, a reporting application was developed to automatically pull data from ConnectWise and generate Excel reports with interactive dashboards, enabling users to filter by company, date range, and report type. These solutions significantly reduced manual workload, improved accuracy, and streamlined internal workflows across the organisation.

Key Tasks:

1. Integrated Vimeo API and Metropolitan Cemetery Board (MCB) API using Python to automate the retrieval of funeral service streaming links based on the deceased's name.
2. Designed and built a desktop user interface using Tkinter to provide staff with a simple, efficient search and retrieval workflow.

3. Developed and implemented a web-based reporting application using HTML, CSS, and JavaScript, enabling users to generate ConnectWise-based Excel reports with selectable filters for company, date range, and report type.
4. Built a Flask-based backend service that integrates with the ConnectWise API to collect structured data and automatically generate Excel reports, supporting both internal dashboards and web-based reporting tools.

Outcome: The project was delivered successfully and received strong appreciation from both managers and team members, as it significantly improved their daily workflows and eliminated time-consuming manual tasks. The solution was particularly valued because it addressed a clear operational gap that I identified shortly after joining the company, demonstrating initiative and a proactive approach to improving internal processes. By automating key functions and creating intuitive tools, the project enhanced efficiency, accuracy, and overall team productivity.

PROJECT NAME: Remittance App (Data Engineering PTY LTD)

Objective: To develop a user-friendly website for a client that needing a platform for their business which is money transfer application. The application itself built using a custom module of Odoo version 17. The website let customer to signup, login, check history of the transactions, storing bank details, creating contact list, and sending money to the contact list. The application itself requires multiple APIs that needed to verify user's ID, Banking services, google autocomplete address, and SMS service. All of these features integrated by creating a custom module in Odoo that written in Python and the front-end is using HTML, SCSS, and JavaScript that stored in XML format.

Key Tasks:

1. Designing a UI/UX for required web page using Figma.
2. Improving the current web page by giving ideas of what can be done and implement the changes.
3. Integrating required APIs to the backend of the website such as Strive, GoogleAPI, Monoova, and Twilio.
4. Develop and improve a responsive web page based on the Figma design using HTML, SCSS, Bootstrap, and JavaScript.
5. Developing database model that is integrated with Odoo using PostgreSQL.

Outcome: The project is still ongoing and being developed based on the requirements and manager's feedback. Currently, the backend processes such as APIs integration are mainly completed and frontend also 80% completed. The website contains multiple pages such as homepage, about page, user dashboard, transactions page, and more. This website are being developed as a custom module that later can be integrated with other Odoo modules to streamline processes.